USAF ACADEMY
DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING
LESSON 31 LAB - Micro-Doppler Classification and Dismount Tracking
Lab Packet
Fall 2026

Name: _____ **Section:** _____ **Date:** _____

Learning Objective 4.8: I can integrate FMCW radar processing with a phased-array front-end to track a moving target.

Learning Objective 4.9: I can classify a detected target from its micro-Doppler signature and report the classifier's accuracy.

This packet is the L31 Lab, in three parts, and you turn it in once, complete, after the lab and before you write the final report. **Part 1, Predict**, is yours to do before the lab: the shaded Predicted cells of each table. **Part 2, Measure**, is your readings and the notebook's in the white Measured cells of the same tables. **Part 3, Analyze**, is yours to write after the lab. Mark the rows that come from recorded data.

Grading. The packet is checked and returned with feedback; it is scaffolding for your final report, not part of it. I read each Part 3 answer for four things:

- a claim: what agrees, what does not, and why;
- the evidence: your own numbers from Part 2, set against your predictions;
- the reasoning that connects the evidence to the claim;
- the uncertainty: how large a difference your measurement can resolve.

Part 1: Predict (individual, before the lab)

1. **Predict.** Fill in every shaded **Predicted** cell of Tables A to E. Lesson 31's Pre-Lab frame names the frame that works each method, on an example burst. Attach your work for one row of each table on your own paper.

    - **The radar.** The Phaser app's Range + Velocity preset: 300 μs chirps of 500 MHz, one every 0.5 ms, at 9.9 GHz, 256 chirps a frame. Velocity is positive closing.
    - **The targets.** A desk fan 15 cm across at 1000 rpm, its blades' weight spread evenly from hub to tip; a fidget spinner 7 cm across at 1200 rpm, its weight at its rim; both edge-on. A hand waving 0.20 m each way at 1.5 Hz. A walker whose torso closes at 1.3 m/s, with limbs that swing up to 2.7 m/s about it. If your own fan or spinner differs, predict for yours and say so.
    - **The walk.** The array is at the origin, y straight out and x to the side, positive on the E8 side, where angles are positive. Mark A is at $x = -0.31$ m, $y = 1.47$ m; mark B at $x = 0.69$ m, $y = 3.94$ m. The walker goes from A to B at 1.0 m/s, turns, and comes back; in the turn the speed swings from $+1$ to -1 m/s over 1 s, as a cosine. The track updates once a frame.
    - **The window.** The spectrogram windows are 16, 64, and 256 chirps.

Part 2: Measure (your kit and the notebook)

Follow the procedure on the L31 Lab page and fill in every white **Measured** cell. Speeds and velocities are in m/s, positive closing; ranges in m; angles in degrees, positive on the E8 side.

2. **Table A: four targets** (Steps 5 to 8), MTI on: the largest speed on each side of 0 m/s, and whether the band is centered.

	Largest −		**Largest +**		**Centered?**	
	Predicted	**Measured**	**Predicted**	**Measured**	**Predicted**	**Meas.**
Fan						
Spinner						
Hand						
Walker						

3. **Table B: the crossing walker** (Steps 9 to 12), MTI on. At the turn, record the torso.

	Range (m)		**Velocity or angle**	
	Predicted	**Measured**	**Predicted**	**Measured**
Mark A, by tape				
Mark B, by tape				
Just after A, going out				
At B, the turn (torso)				
Just before A, coming back				
Angle at B (angle cell)	–	–		

4. **Table C: the window** (Step 14), on the spinner's and the walker's record 1.

	Resolution	**Torso line**	**Spinner line**	**Walker spread**
`N_WIN`	**Predicted**	**Measured**	**Measured**	**Measured**
16				
64				
256				

5. **Table D: the classifier** (Step 15), each record scored by a classifier trained on the other 15. Rows are the true class; columns, the call.

	Three features				+ edge				+ shape			
true	W	F	S	H	W	F	S	H	W	F	S	H
walker												
fan												
spinner												
hand												
accuracy												

Shape, spread over edge	**Predicted**	**Measured**
Fans (mean of four)		
Spinners (mean of four)		

Records your best feature set still calls wrong:

6. **Table E: the track** (Step 16) of the recorded walk, scored against its truth after the first second. The noise ratio is the track's range RMS over the detections'. Errors are in m and degrees.

	Range RMS	**Noise ratio**		**Range error at the turn**		**Angle RMS**
	Measured	**Predicted**	**Measured**	**Predicted**	**Measured**	**Measured**
Detections alone		–	–	–	–	
$\alpha = 0.2$						
$\alpha = 0.5$						
$\alpha = 0.8$						
Yours, α =		–	–	–		

Part 3: Analyze (individual, after the lab)

Answer each prompt from your own Part 2 readings and your own Part 1 predictions, and keep the notebook's printouts beside you.

7. **Speeds and windows.** Compare your Table A speeds and your Table C line widths with your predictions. Say which differences your readings can resolve, give the cause you think most likely for each one they can, and say what you would change in the reading or the processing to test it.

8. **What told the targets apart.** Compare your three confusion matrices in Table D, with the features the notebook printed for each record. Explain what moved each record that changed between the matrices, give the cause you think most likely for every record your best set still calls wrong, and describe a change to the records or the features that would test it.

9. **The track.** Compare your Table E rows with your predictions, in range and in angle. Choose the gains you would run on this radar for each, and support the choice with your own numbers and how large a difference they can resolve.

Documentation: _____